

Chapter 79 — Neoplasms of the Testis — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Solid testicular mass = germ cell tumour (GCT) until proven otherwise** → radical inguinal orchidectomy.
- **Trans-scrotal orchidectomy/biopsy contraindicated** — alters lymphatic drainage.
- **Raised alpha-fetoprotein (AFP) = non-seminoma (NSGCT)**, even if histology says pure seminoma.
- **Half-lives: AFP 5–7 days; hCG (human chorionic gonadotrophin) 24–36 h.**
- **Rising post-orchidectomy markers** → induction chemotherapy.
- **International Germ Cell Cancer Collaborative Group (IGCCCG):** good → BEP×3 (bleomycin, etoposide, cisplatin) or EP×4 (etoposide, cisplatin); intermediate/poor → BEP×4.
- **CS I NSGCT (clinical stage I) occult metastasis 20–30%;** predictors: lymphovascular invasion (LVI), embryonal carcinoma (EC) predominance.
- **CS I seminoma → surveillance preferred;** risk factors: size >4 cm, rete testis invasion.
- **NSGCT residual mass >1 cm → resect;** teratoma ≈45%, viable ≈15%.
- **Seminoma residual >3 cm → FDG-PET (fluorodeoxyglucose positron emission tomography) ≥6 weeks;** PET+ → resect; <3 cm or PET- → observe.

EPIDEMIOLOGY AND RISK

- **Commonest:** solid malignancy, males **20–40**; incidence peak **25–35**.
- **GCT >95%** of testis tumours; **5% extragonadal**.
- **Seminoma 52–56% vs NSGCT 44–48%.**
- **Bilateral 2%,** mostly metachronous; synchronous 0.5%.
- **Cryptorchidism relative risk (RR) 4–6 → 2–3** if orchidopexy before puberty.
- **Affected brother RR 8–12 vs father 2–4.**
- **Prior GCT:** contralateral risk ×12; 15-year incidence only 2%.
- **GCNIS (germ cell neoplasia in situ):** invasive GCT 50% at 5 years, 70% at 7.
- **Contralateral GCNIS 5–9%;** ≈36% if atrophy/cryptorchidism/age <40.
- **Metastatic cure:** 5–10% pre-cisplatin → 80–90% now.

HISTOLOGY

- **i(12p)** in all postpubertal GCT except spermatocytic tumour.
- **Any NSGCT element** → classify as NSGCT.
- **Seminoma:** CD117+, CD30-, placental alkaline phosphatase (PLAP) strongly +; hCG in 15%.
- **Spermatocytic tumour:** sixth decade, no GCNIS, benign; orchidectomy curative.
- **EC:** c-KIT-; metastasises with normal markers.
- **Yolk sac:** AFP, Schiller–Duval body.
- **Choriocarcinoma:** very high hCG, haematogenous, haemorrhage; thyrotoxicosis (shared α-subunit).
- **Teratoma:** chemoresistant; in ≈half of mixed GCT; malignant transformation → rhabdomyosarcoma commonest.

PRESENTATION AND DIAGNOSIS

- **Metastatic at diagnosis:** ≈two-thirds NSGCT vs 15% seminoma.
- **Misdiagnosed initially ≈one-third** (epididymitis, hydrocele).
- **Suspect GCT, age 15–50:** testicular mass, midline retroperitoneal mass, left supraclavicular mass.
- **Epididymo-orchitis:** re-examine 2–4 weeks after antibiotics.
- **Ultrasound first-line;** microlithiasis alone → no action.
- **Impalpable lesion:** <1 cm ≈50% malignant; 1–2 cm ≈80%.
- **Orchidectomy:** within 1–2 weeks; cord to internal ring.
- **Orchidectomy alone cures:** 80–85% CS I seminoma; 70–80% CS I NSGCT.
- **Testis-sparing:** only bilateral/solitary testis, tumour <2–3 cm; frozen section.
- **GCNIS dose:** radiotherapy ≥20 Gy (<20 Gy → recurrence); up to 40% need testosterone.

- **hCG >5000 IU/L** → usually NSGCT.
- **False-positive hCG:** luteinising hormone (LH) cross-reactivity (hypogonadism) → normalises 48–72 h after testosterone.
- **miR-371a-3p:** best microRNA marker; **negative in teratoma.**
- **Unstable, markers very high** → chemotherapy first, orchidectomy after (sanctuary site).
- **Extragonadal pattern + raised AFP/hCG** → treat without biopsy.

STAGING

- **Landing zone:** right → interaortocaval; left → para-aortic (PA)/left renal hilar.
- **Crossover** right → left common; left → right implies bulk.
- **Computed tomography (CT) false-negative:** 25–35% CS I NSGCT (1-cm cut-off).
- **Suspicious node:** 5–9 mm in landing zone, anterior to great vessels.
- **CS II:** IIA ≤2 cm; IIB 2–5 cm; IIC >5 cm.
- **pN:** pN1 ≤5 nodes ≤2 cm; pN2 >5 nodes or 2–5 cm; pN3 >5 cm.
- **S category:** S1 AFP <1000, hCG <5000, lactate dehydrogenase (LDH) <1.5×N (upper limit of normal); S3 AFP >10,000, hCG >50,000, LDH >10×N.
- **LVI** → pT2.
- **Imaging and markers:** within 4 weeks of treatment; equivocal CT → repeat 6–8 weeks.
- **Brain imaging if hCG >10,000;** no routine bone scan or diagnostic PET.
- **IGCCCG NSGCT good:** AFP <1000, hCG <5000, LDH <1.5×N, no NPVM (non-pulmonary visceral metastases); 56%, 5-year survival 92%.
- **IGCCCG NSGCT poor:** mediastinal primary, NPVM, AFP >10,000, hCG >50,000, LDH >10×N; 16%, survival 48%.
- **IGCCCG seminoma:** NPVM only factor; 90% good; **no poor risk.**
- **Fertility:** recovery after chemotherapy 50% at 2 years, 80% at 5; bank sperm before RPLND/chemotherapy/radiotherapy.

NON-SEMINOMA

- **CS I options:** surveillance, RPLND (retroperitoneal lymph node dissection), BEP×1 — all ≈100% survival.
- **No LVI/EC:** occult metastasis <20%.
- **Surveillance:** relapse ≈28%; >90% within 2 years; 65–75% retroperitoneal.
- **Surveillance best if relapse risk <33–37%;** active treatment >46–54%.
- **Full bilateral template:** recurrence <2%; antegrade ejaculation >90% (nerve-sparing).

- **Albers trial:** BEP×1 vs unilateral RPLND → 2-year progression-free survival 99% vs 92%; no deaths.
- **pN2-3 relapse >50%** → adjuvant BEP×2/EP×2 → ≤1%.
- **Primary BEP×1:** relapse ≈2-3%, but relapses chemoresistant.
- **CS IS** → induction chemotherapy.
- **CS IIA-B chemotherapy if:** raised markers or nodes >3 cm.
- **CS IIA-B RPLND if:** normal markers, nodes <3 cm (teratoma risk).
- **Gold standard first line:** BEP; PVB (cisplatin, vinblastine, bleomycin) replaced (neuromuscular toxicity).
- **VIP×4 (etoposide, ifosfamide, cisplatin)** replaces BEP×4 when lungs compromised/major chest surgery.
- **Complete response:** residual ≤1 cm; residual ≤20 mm still teratoma ≈30%.
- **Resect RPLND first;** necrosis there predicts necrosis elsewhere.
- **Post-chemotherapy viable cancer:** bad if IGCCCG intermediate/poor, incomplete resection, >10% viable.
- **Second line:** VIP, VeIP, TIP (ifosfamide-cisplatin with etoposide, vinblastine or paclitaxel); HDCT (high-dose chemotherapy) specialist centres only.
- **Growing teratoma syndrome:** falling markers + enlarging mass → stop chemotherapy, resect.
- **Late relapse >2 years (≈3%):** yolk sac commonest; surgery; TIP best chemotherapy.
- **Desperation surgery:** rising markers, single resectable site → survival 33-57%.

SEMINOMA

- **CS I ≈80%;** occult metastasis 10-15%; surveillance relapse-free 80-86%.
- **Seminoma relapse later:** median 14 months vs 4-8 (NSGCT).
- **CT detects >85% of relapses;** hCG <5%.
- **PA radiotherapy 20 Gy = 30 Gy;** PA → pelvic recurrence 2% → pelvic CT.
- **MRC TE19/EORTC 30982 (Medical Research Council / European Organisation for Research and Treatment of Cancer):** carboplatin×1 vs radiotherapy, relapse-free 94.7% vs 96%.
- **Carboplatin dose = $7 \times (\text{glomerular filtration rate [GFR]} + 25)$ mg;** isotope GFR, not estimated.
- **Non-compliant:** radiotherapy or carboplatin×2.

- **CS IIA-B:** dog-leg (DL) radiotherapy 25-30 Gy (boost 5-10 Gy), BEP×3/EP×4, or RPLND ≤3 cm.
- **Bulky >3 cm/multiple** → chemotherapy.
- **SEMS:** RPLND, 2-year recurrence-free 81%, survival 100%.
- **Advanced:** 90% good → BEP×3/EP×4; intermediate → BEP×4.
- **Residual histology:** necrosis 90%; viable in 13-55% if >3 cm vs 0-4% <3 cm.
- **PET >3 cm:** specificity 100%, sensitivity 80%.
- **Relapse after carboplatin** = chemo-naïve → cisplatin-based first line.

BRAIN METASTASES AND TOXICITY

- **Brain metastases:** 1% at diagnosis; choriocarcinoma; haemorrhage deaths 4-10%.
- **At diagnosis:** BEP×4 → resect residual.
- **5-year survival:** NSGCT 33%, seminoma 57%.
- **Toxic death:** 0-2.4% good risk; 3-4.4% intermediate/poor.
- **SMN (second malignancy):** +60-100% after cisplatin or radiotherapy; +200% after both.
- **After radiotherapy:** SMN ≈18% at 25 years.

CLASSIC TRAPS

- **Trap:** normal markers exclude GCT — **Actually:** orchiectomy regardless.
- **Trap:** pre-orchidectomy markers stage disease — **Actually:** post-orchidectomy values.
- **Trap:** IGCCCG for relapse — **Actually:** diagnosis of metastatic disease only.
- **Trap:** LVI predicts seminoma relapse — **Actually:** size >4 cm, rete testis.
- **Trap:** spermatocytic tumour is a seminoma — **Actually:** distinct, benign, no GCNIS.
- **Trap:** PET for NSGCT residual — **Actually:** useless; teratoma not FDG-avid.
- **Trap:** no teratoma in primary excludes it — **Actually:** does not; resect.
- **Trap:** carboplatin for metastatic GCT — **Actually:** inferior to cisplatin.
- **Trap:** mediastinal seminoma is poor risk — **Actually:** seminoma has no poor risk.
- **Trap:** high-dose chemotherapy is standard — **Actually:** three negative randomised trials.